Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for Airborne vs. Mechanical Low-Frequency Excitation

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- Fluid-filled shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at 2900 Hz; flexural modes match ISO 2631 data
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Mechanical vibration couples 1000–10000x more efficiently
- Coupling disparity explains WBV effects vs. absent airborne effects

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for Airborne vs. Mechanical Low-Frequency Excitation

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Abstract

The human abdomen is modelled as a fluid-filled viscoelastic oblate spheroidal shell to investigate its modal response under low-frequency excitation. Using Love–Kirchhoff thin-shell theory with Kelvin–Voigt viscoelasticity and proper fluid–structure coupling, we derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families. The breathing mode, dominated by the fluid bulk modulus, occurs at approximately 2900 Hz and is irrelevant to infrasonic exposure. In contrast, flexural modes—in which the shell deforms without significant volume change and the enclosed fluid acts primarily as added mass—fall in the 4 Hz to 10 Hz range, consistent with abdominal resonance data reported under ISO 2631. A comprehensive parametric study over elastic modulus (0.05 MPa to 2.0 MPa), geometry, wall thickness, and intra-abdominal pressure demonstrates that the $n = 2$ flexural mode robustly occupies this frequency band for relaxed musculature, though the precise frequency is sensitive to tissue stiffness and boundary conditions.

We then compare two excitation pathways quantitatively. For airborne acoustic coupling, the long-wavelength condition ($ka \approx 0.017$ at 7 Hz) imposes a $(ka)^n$ penalty on flexural mode excitation, yielding displacements of order 0.1 µm at 120 dB SPL—below known mechanotransduction thresholds. For mechanical (whole-body) vibration, coupling is direct and produces displacements 10^3–10^4 times larger at occupational exposure levels. This disparity explains why whole-body vibration at 4 Hz to 8 Hz is associated with gastrointestinal

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effects in the epidemiological literature, whereas airborne infrasound at comparable frequencies is not. The analysis provides the first unified quantitative framework bridging these two excitation mechanisms through a common modal description.

Keywords: fluid-filled shell vibration, viscoelastic oblate spheroid, infrasound, whole-body vibration, acoustic–structural coupling, ISO 2631

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1. Introduction

Low-frequency acoustic and vibrational exposures are ubiquitous in occupational and environmental settings—from wind turbines and heavy machinery to transport vehicles and industrial plant. The human body is known to exhibit resonance phenomena in the 4 Hz to 8 Hz range under whole-body vibration, with the abdominal viscera identified as the primary resonant subsystem [1, 2].

A persistent claim in popular culture holds that a specific infrasonic frequency in the 5 Hz to 9 Hz range—the so-called "brown note"—can cause involuntary loss of bowel control through resonance of the abdominal cavity. Despite widespread repetition, no rigorous scientific study has confirmed this claim, and it has been explicitly debunked in controlled experiments [3].

However, the complete absence of a *quantitative mechanistic analysis* leaves the question incompletely resolved. No study has:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways using a common modal framework.

The present work addresses all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell and derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families. We then

compare the efficiency of airborne acoustic coupling (subject to a $(ka)^n$ penalty) against mechanical whole-body vibration coupling (which bypasses this penalty entirely).

The central finding is that the coupling disparity between these two pathways—a factor of 10^3–10^4—quantitatively explains why whole-body vibration at 4 Hz to 8 Hz produces gastrointestinal effects while airborne infrasound at comparable frequencies does not.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation for the fluid-filled viscoelastic shell, including both breathing and flexural mode families and the two coupling pathways. Section 3 presents a comprehensive parametric study and validates the model against ISO 2631 data. Section 4 quantifies the airborne–mechanical coupling disparity. Section 5 interprets the results and discusses limitations, and Section 6 summarises the conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [4, 5], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 160\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is therefore applicable.

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Relaxed passive tissue
Poisson's ratio	ν	0.45	—	Nearly incompressible
Wall density	ρ_w	1050	$\mathrm{kg/m^3}$	Soft tissue
Fluid density	ρ_f	1040	$\mathrm{kg/m^3}$	Water-like
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Resting supine
Loss tangent	η	0.30	—	Viscoelastic tissue

2.2. Shell Constitutive Model

The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [6], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.30$ ($Q = 3.3$) represents a central estimate.

2.3. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\text{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit area is

$$k_{\text{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\text{GPa}$ and $R = 0.157\,\text{m}$, $k_{\text{fluid}} = 4.2 \times 10^{10}\,\text{Pa/m}$, which exceeds the membrane stiffness ($k_{\text{shell}} \approx 2.4 \times 10^5\,\text{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\text{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2900\ \text{Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.4. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

Following Lamb [5] and Junger & Feit [4], the flexural mode of order n has three contributions to the restoring force per unit area:

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.4\,\text{Hz}$, $f_3 = 6.7\,\text{Hz}$, and $f_4 = 9.4\,\text{Hz}$. These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [7] and Mansfield [8].

2.5. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At 7 Hz, the acoustic wavelength in air is $\lambda \approx 49\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f R}{c_{\text{air}}} \approx 0.017 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales as [4]:

$$p_{\text{eff}}^{(n)} = p_{\text{inc}} \times (ka)^n. \tag{15}$$

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 2.8 \times 10^{-4}. \tag{16}$$

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$), equation (17) gives $\xi_{\text{air}} \approx 0.14\,\mu\text{m}$ for $n = 2$ — well below the 0.5–2.0 μm activation threshold of mechanosensitive ion channels [9].

2.6. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no impedance mismatch and no $(ka)^n$ penalty. The vibration is transmitted at full amplitude through the rigid skeletal frame.

For harmonic base acceleration $a_{\text{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\text{base}} = \frac{a_{\text{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\text{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\text{mech}} = x_{\text{base}} \times H_{\text{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\mathrm{rms}} = 1.15\,\mathrm{m/s^2}$ and at the $n = 2$ resonance frequency:

$$\xi_{\mathrm{mech}} \approx 7{,}000\ \mu\mathrm{m} = 7\ \mathrm{mm}, \tag{21}$$

which exceeds the mechanotransduction threshold by more than three orders of magnitude.

2.7. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure of the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \sim 10^3\text{–}10^4. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and is *robust* to uncertainties in material properties and boundary conditions because both pathways excite the same modal structure. The coupling ratio is the central result of this work: it provides a quantitative explanation for why whole-body vibration at 4–8 Hz is associated with gastrointestinal effects in the epidemiological literature [2], whereas airborne infrasound at comparable frequencies is not.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range.

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5\,\mathrm{MPa}$, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range.

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$).

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.6	Marginal
0.10	4.4	Yes
0.20	5.6	Yes
0.50	8.3	Yes
1.00	11.5	No
2.00	16.1	No

3.2. Multi-Parameter Sensitivity

A full factorial analysis over 486 parameter combinations (6 values of E, 3 semi-major axes, 3 aspect ratios, 3 wall thicknesses, 3 loss tangents) reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in the broader 4 Hz to 8 Hz ISO 2631 range. The dominant parameter is E, followed by geometry and intra-abdominal pressure.

3.3. Multi-Layer Wall Model

The abdominal wall comprises approximately five distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Using classical laminate theory, the effective composite modulus for a relaxed wall is $E_{\mathrm{eff}} \approx 0.075\,\mathrm{MPa}$ ($h_{\mathrm{total}} = 25.1\,\mathrm{mm}$), giving $f_2 = 4.6\,\mathrm{Hz}$—within 4% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\mathrm{eff}} = 2.7\,\mathrm{MPa}$) shifts f_2 to 23.1 Hz.

3.4. Boundary Condition Effects

The free-shell model provides a lower bound on frequency. Partial constraints (spine, pelvis, ribs) raise the $n = 2$ frequency by an estimated factor of 1.3–3.0×, depending on the angular extent of the constrained boundary. Even with a 2×

constraint multiplier, the mode remains within the ISO 2631 range for soft tissue.

3.5. ISO 2631 Validation

Table 3 compares model predictions against published abdominal resonance data.

Table 3: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.3	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.1	Marginal

4. Airborne vs. Mechanical Coupling

This section presents the central novel result of this work: a quantitative comparison of airborne acoustic and mechanical (whole-body vibration) coupling to the abdominal flexural modes derived in §2.

4.1. Airborne Acoustic Pathway

Table 4 summarises the predicted airborne displacement at resonance for the $n = 2$ flexural mode across a range of sound pressure levels.

Table 4: Airborne displacement at $n = 2$ resonance ($f_2 = 4.4\,\mathrm{Hz}$, $Q = 3.3$, $E = 0.1\,\mathrm{MPa}$).

SPL (dB)	p_{inc} (Pa)	p_{eff} (Pa)	ξ (μm)	PIEZO?
100	2.0	3.2×10^{-4}	0.014	No
110	6.3	1.0×10^{-3}	0.045	No
120	20.0	3.2×10^{-3}	0.14	No
130	63.2	1.0×10^{-2}	0.45	No
140	200.0	3.2×10^{-2}	1.44	Marginal

At 120 dB SPL (the threshold of pain), the predicted wall displacement is 0.14 µm—well below the 0.5–2.0 µm activation range of mechanosensitive PIEZO channels. Reaching the 1 µm threshold requires approximately 137 dB, a level associated with close proximity to jet engines and acoustic weapons.

4.2. Mechanical (Whole-Body Vibration) Pathway

In contrast, mechanical vibration transmitted through the skeleton couples directly to the shell without the $(ka)^n$ penalty. Table 5 shows the predicted relative displacement for the $n = 2$ mode under whole-body vibration.

Table 5: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.4\,\mathrm{Hz}$, $Q = 3.3$).

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	PIEZO?
0.1	187	623	Yes
0.5 (EU action)	935	3115	Yes
1.0	1869	6230	Yes
1.15 (EU limit)	2150	7165	Yes

At the EU Whole-Body Vibration Directive action value of 0.5 m/s^2, the predicted displacement is 3115 µm—exceeding the PIEZO threshold by more than three orders of magnitude.

4.3. Coupling Ratio

The ratio of mechanical to airborne displacement defines the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \sim 10^3\text{–}10^4. \tag{23}$$

This ratio depends primarily on $(ka)^{-n}$ and the base-excitation amplification H_{rel}, and is robust to uncertainties in material properties because both pathways excite the same modes.

4.4. Energy Budget

To verify self-consistency, we compute the absorption cross-section using reciprocity:

$$\sigma_{\text{abs}} = \frac{(2n+1)\lambda^2}{4\pi} \frac{\Gamma_{\text{rad}}}{\Gamma_{\text{total}}}, \tag{24}$$

where Γ_{rad} is the radiation damping width and $\Gamma_{\text{total}} = \Gamma_{\text{rad}} + \Gamma_{\text{struct}}$. For the $n = 2$ mode in air, $\zeta_{\text{rad}} \approx 10^{-15}$ whereas $\zeta_{\text{struct}} = 0.15$, giving an absorption efficiency of order 10^{-14}. The shell is effectively transparent to airborne infrasound.

4.5. Comparison with Empirical Evidence

The model predictions are consistent with two independent bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1].
- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL ≤ 150 dB [3].

The coupling disparity provides a quantitative explanation for this apparent contradiction: the two bodies of literature address different excitation mechanisms with fundamentally different coupling efficiencies.

5. Discussion

5.1. The Coupling Disparity Resolves a Paradox

The central result of this study is the 10^3–10^4 factor separating mechanical from airborne coupling efficiency. This disparity has a simple physical origin: mechanical vibration drives the shell wall directly, while airborne sound must couple through the acoustic radiation impedance mismatch. In the long-wavelength limit ($ka \ll 1$), this mismatch imposes a $(ka)^n$ penalty on flexural mode n, which at 7 Hz amounts to a factor of 2.8×10^{-4} for $n = 2$.

This finding resolves an apparent paradox in the literature: whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress (including nausea, abdominal pain, and in extreme cases, incontinence) [2, 8], whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [3].The modal analysis shows that both observations are consistent with the same underlying resonance—but the two excitation pathways differ by three to four orders of magnitude in coupling efficiency.

5.2. Implications for the "Brown Note" Hypothesis

The results indicate that abdominal resonance in the 4 Hz to 10 Hz range is physically real and well-supported by ISO 2631 data. The "brown note" hypothesis is therefore half-correct: the resonance exists. However, *airborne* excitation of this resonance is energetically negligible at any plausible SPL. The effects attributed to infrasound are instead explained by mechanical vibration, which is the actual exposure pathway in the occupational scenarios (heavy equipment, transport vehicles) where gastrointestinal symptoms are reported.

5.3. Gas Pocket Mechanism

We note that bowel gas pockets (radius $R \approx 2\,\mathrm{cm}$) have Minnaert resonance frequencies in the range 30 Hz to 330 Hz. At infrasonic frequencies, sub-resonant but non-negligible acoustic displacements of order 1 µm at 120 dB may occur within these pockets. This mechanism could potentially explain inter-individual variability in infrasound sensitivity, but the present analysis cannot make a strong quantitative claim without modelling the gas–tissue interaction in detail.

5.4. Limitations

1. **Boundary conditions**: The free-shell model provides a lower bound on mode frequencies. Partial clamping by the spine and pelvis will raise the $n = 2$ frequency, potentially by a factor of 2–3×. Finite element validation is needed to resolve this uncertainty.
2. **Linear theory**: The small-displacement assumption may not hold for large-amplitude mechanical vibration.

3. **Homogeneous wall**: The abdominal wall is a multi-layer composite, though our laminate analysis shows this introduces less than 5% error in the eigenfrequency.
4. **Thin-shell theory**: The wall-thickness-to-radius ratio ($h/R \approx 0.06$) is near the upper bound of thin-shell validity.
5. **No organ-level model**: Internal organs are treated as fluid; a more realistic model would include solid inclusions.

6. Conclusions

A fluid-filled viscoelastic oblate spheroidal shell model of the human abdomen reveals two distinct mode families with fundamentally different physics. The breathing mode ($n = 0$), governed by the fluid bulk modulus, occurs at approximately 2900 Hz and is irrelevant to infrasonic exposure. Flexural modes ($n \geq 2$), in which the fluid acts only as added mass, fall in the 4 Hz to 10 Hz range for relaxed musculature, consistent with ISO 2631 whole-body vibration data.

The central finding is a 10^3–10^4 coupling disparity between mechanical and airborne excitation of these flexural modes. This disparity, arising from the $(ka)^n$ long-wavelength penalty on acoustic coupling, provides a quantitative mechanistic explanation for an otherwise puzzling asymmetry in the empirical literature: mechanical vibration at 4 Hz to 8 Hz produces gastrointestinal effects, while airborne infrasound at comparable frequencies does not. The "brown note" hypothesis is thus physically grounded—the abdominal resonance is real—but the postulated *airborne* excitation pathway is energetically negligible.

Future work should validate the boundary condition assumptions through finite element analysis, extend the model to include gas pocket resonance in the bowel, and investigate the non-linear regime relevant to high-amplitude mechanical vibration.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jon: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.